

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)



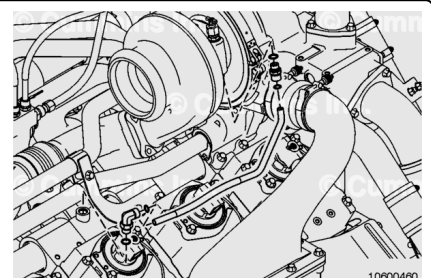
Remove

Single Stage

Note : The following steps for the coolant supply and return hoses apply to all four high pressure turbocharger locations.

With Mechanically Actuated Injector

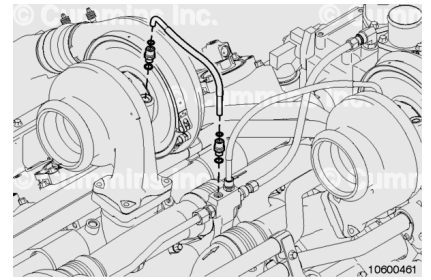
- Disconnect the coolant supply hose from the turbocharger and the water manifold.
- Remove the male union from the turbocharger and the elbow from the water manifold.
- Discard the o-rings.



Disconnect the coolant return hose from the turbocharger and the water transfer connection.

Remove the male union from the turbocharger and water transfer connection.

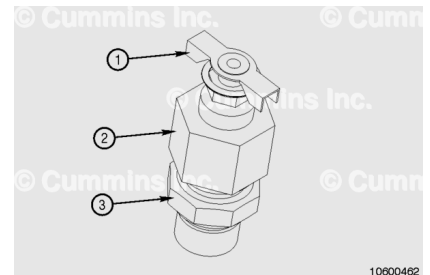
Discard the o-rings.



Note : The following procedure to remove the air bleed petcock is the same for the right front and the left rear high pressure turbocharger locations.

1. Remove the petcock from the nut.
2. Remove the nut from the union.
3. Remove the union from the turbocharger.

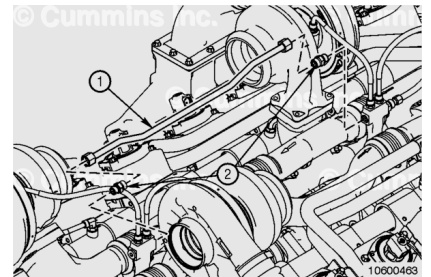
Discard the o-ring.



Remove the coolant return hose (1) from both the water transfer connections.

Remove both unions (2).

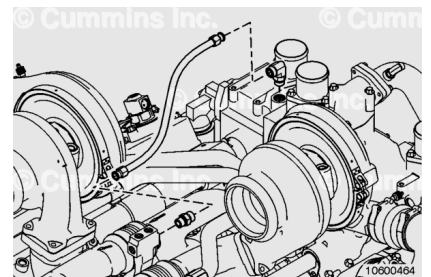
Discard the o-rings.



Remove the coolant return hose from the water transfer connection and the thermostat housing.

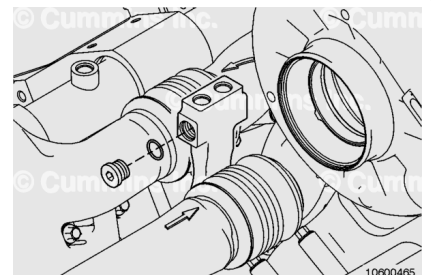
Remove the union from the water transfer connection and the elbow from the thermostat housing.

Discard the o-rings.



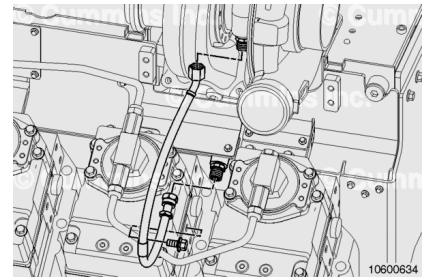
Remove the plug from the water transfer connection.

Discard the o-ring.



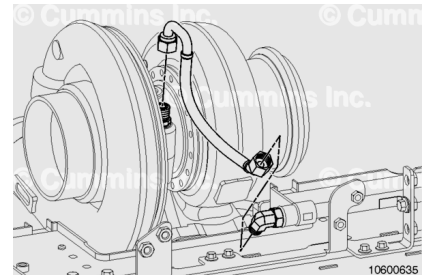
With Electronically Actuated Injector

- Disconnect the coolant supply hose from the turbocharger and the water manifold.
- Discard the o-rings.



Disconnect the coolant return hose from the turbocharger and the water transfer connection.

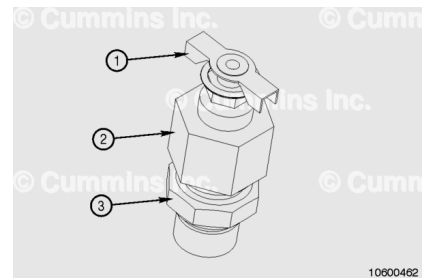
Discard the o-rings.



Note : The following procedure to remove the air bleed petcock is the same for the right front and the left rear high pressure turbocharger locations.

1. Remove the petcock from the nut.
2. Remove the nut from the union.
3. Remove the union from the turbocharger.

Discard the o-ring.



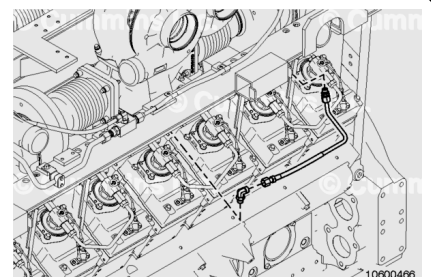
Two Stage

Note : The following steps for the coolant supply and return hoses apply to all four high pressure turbocharger locations with Electronically Actuated Injector.

Disconnect the coolant supply hose from the turbocharger and the water manifold.

Remove the union fittings from the turbocharger and the elbow fitting from the water manifold.

Discard the o-rings.

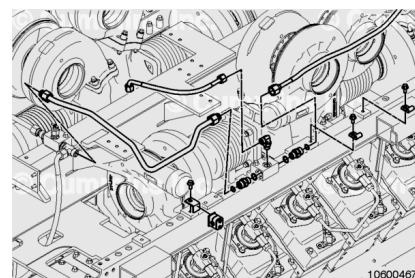


For the rear turbochargers, remove the p-clips securing the rear turbocharger coolant lines to the two stage frame.

Remove the turbocharger coolant return lines from the turbocharger and the manifold.

For the rear turbochargers, remove the union from the turbocharger and the reducer and union from the manifold. For the front turbochargers, remove the union from the turbocharger and the elbow from the manifold.

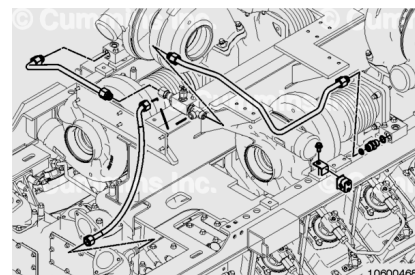
Discard the o-rings.



Remove the clip securing the coolant return line at the left bank front of the engine.

Remove the coolant return line from the side manifold to the front center manifold on each side.

Remove the coolant return line from the center manifold to the thermostat housing.

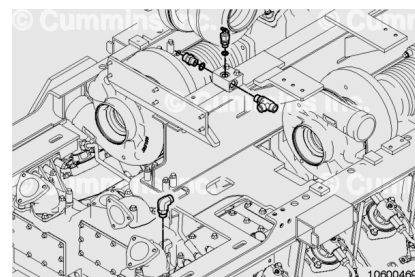


Remove the male unions and T-fitting from the manifolds.

Remove the elbow from the thermostat housing.

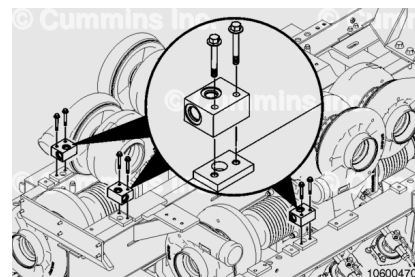
Remove the drain cock from the front center manifold.

Discard the o-rings.



Remove the capscrews securing each of the three manifolds and the two stage frames.

Remove the manifolds.

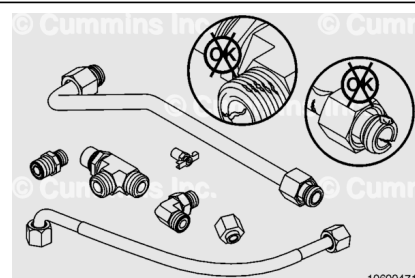


Preparatory Steps

Check the coolant supply and return hose for cracks.

Inspect the unions, elbows, T-fittings, reducer nuts, and drain cocks for stripped threads, cracks, and other damage.

Replace if damage is found.

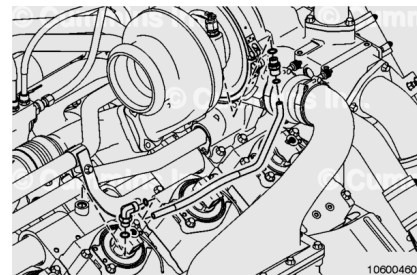


Install

Single Stage

⚠ CAUTION ⚠

When installing flexible hoses be sure that all the original clips are fitted and that the hoses do not contact any other hoses or engine components. Interference between hoses and other engine components can lead to leakages and engine damage.



Note : The following steps are for the coolant supply and return hoses and apply to all four high pressure turbocharger locations.

With Mechanically Actuated Injector

- Install the new o-ring on the supply and return union fittings and install the fittings on the turbochargers.

Torque Value:

Turbocharger Fitting 72 n•m [53 ft-lb]

- Install the new o-ring on the elbow fitting and install the fitting into the water manifold.

Torque Value:

Water Manifold Elbow 72 n•m [53 ft-lb]

- Install new face seal o-rings and connect the turbocharger coolant supply hose between the turbocharger union and the water manifold elbow.

Torque Value:

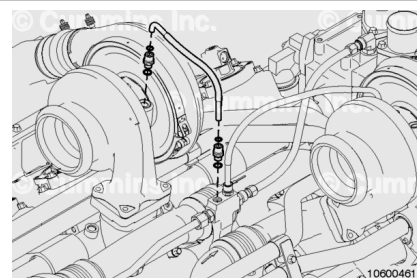
Coolant Supply Hose 45 n•m [33 ft-lb]

Install new o-rings on the male unions to the turbocharger and water transfer connection and install.

Torque Value:

Male Union 72 n•m [53 ft-lb]

Install new face seal o-rings and connect the turbocharger coolant return line between the turbocharger union and the water transfer connection.



Torque Value:

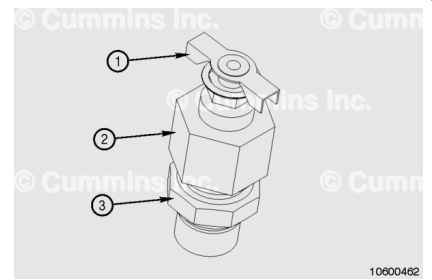
Coolant Supply Hose 45 n•m [33 ft-lb]

Install a new face seal o-ring on the drain cock nut (3) and install the nut.

Torque Value:

Drain Cock Nut 45 n•m [33 ft-lb]

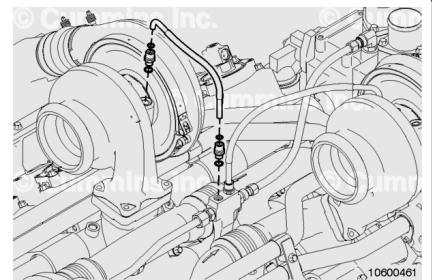
Apply thread sealant, Cummins® Part Number 3375066, or equivalent, to the drain cock (2). Install the drain cock into the nut and tighten two turns from finger tight.



Install new o-rings on the remaining unions and install them into the water transfer connection.

Torque Value:

Water Transfer Connection Union 100 n•m [74 ft-lb]



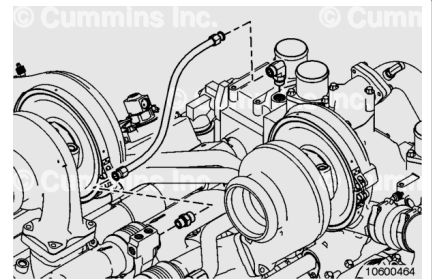
Install a new o-ring on the elbow fitting and install the elbow fitting into the thermostat housing.

Install the new o-ring and install the male union into the water transfer connection.

Torque Value:

Thermostat Housing Elbow and Union 100 n•m [74 ft-lb]

Install a new face seal o-ring and connect the turbocharger coolant return line between the water transfer connection and the thermostat housing elbow.

**Torque Value:**

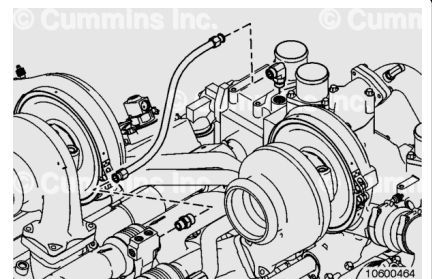
Coolant Return Line 65 n•m [48 ft-lb]

Install a new o-ring on the unions (2) and install them into the water transfer connections.

Torque Value:

Water Transfer Union 72 n•m [53 ft-lb]

Install new face seal o-rings and connect the turbocharger coolant return line between the turbocharger and the water transfer connections.

**Torque Value:**

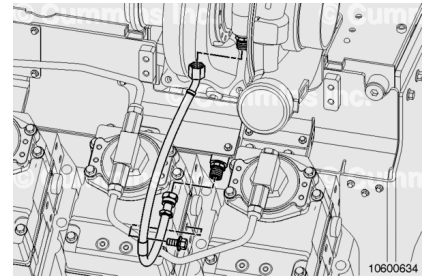
Coolant Return Hose 45 n•m [33 ft-lb]

With Electronically Actuated Injector

- Install new o-rings on the coolant supply hose.
- Connect the coolant supply hose to turbocharger and water manifold.

Torque Value:

Coolant Supply Hose 45 n•m [33 ft-lb]

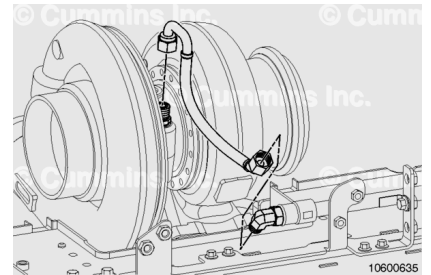


Install new o-rings on the coolant return hose.

Connect the coolant return hose to the turbocharger and the water transfer connection.

Torque Value:

Coolant Return Hose 45 n•m [33 ft-lb]

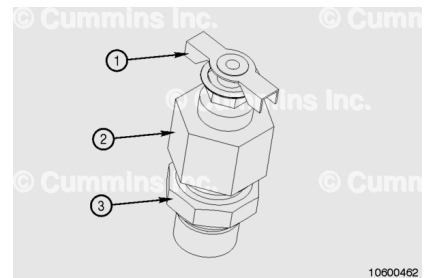


Install a new face seal o-ring on the drain cock nut (3) and install the nut.

Torque Value:

Drain Cock Nut 45 n•m [33 ft-lb]

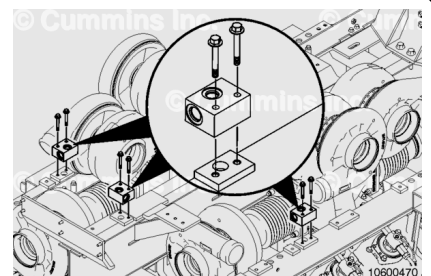
Apply thread sealant, Cummins® Part Number 3375066, or equivalent, to the drain cock (2). Install the drain cock into the nut and tighten two turns from finger tight.



Two Stage

⚠ CAUTION ⚠

When installing flexible hoses be sure that all the original clips are fitted and that the hoses do not contact any other hoses or engine components. Interference between hoses and other engine components can lead to leakages and engine damage.



Install the capscrews securing each of the three manifolds to the stage frame.

Torque Value:

Manifold Capscrew 45 n•m [33 ft-lb]

Install a new o-ring on the elbow and install it in the thermostat housing.

Torque Value:

Thermostat Housing Elbow 100 n•m [74 ft-lb]

Install a new o-ring onto the T-fitting and install it into the front center manifold.

Torque Value:

T-fitting 100 n•m [74 ft-lb]

Install a new o-ring on each of the remaining large unions and install them into the manifolds.

Torque Value:

Large Union 100 n•m [74 ft-lb]

Install a new o-ring onto the drain cock union and install it into the front center manifold.

Torque Value:

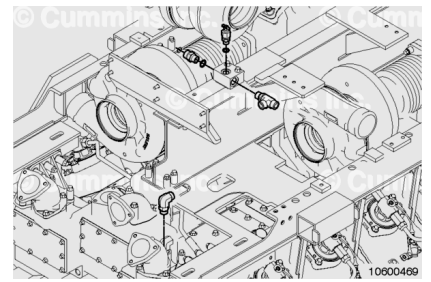
Drain Cock Union 72 n•m [53 ft-lb]

Install a new face seal o-ring on the drain cock nut and install the nut.

Torque Value:

Drain Cock Nut 45 n•m [33 ft-lb]

Apply thread sealant, Cummins® Part Number 3375066, or equivalent, to the drain cock. Install the drain cock into the nut and tighten two turns from finger tight.



Install new face seal o-rings on the coolant return line and install it between the front center manifold T-fitting and the thermostat elbow fitting.

Torque Value:

Coolant Return Line 65 n•m [48 ft-lb]

Install new face seal o-rings on the coolant return line and install it between the front center manifold and the right bank manifold.

Torque Value:

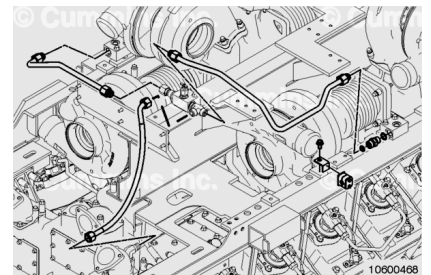
Coolant Return Line Right Bank 65 n•m [48 ft-lb]

Install new face seal o-rings on the coolant return line and install it between the front center manifold and the left bank manifold.

Torque Value:

Coolant Return Line Left Bank 65 n•m [48 ft-lb]

Install the clips securing the coolant return line to the two stage frame.



Torque Value:

Line Clip 45 n•m [33 ft-lb]

Install a new face seal o-ring on each of the reducers. Install the reducers and nuts to the right and left bank manifold unions.

Torque Value:

Reducer 80 n•m [60 ft-lb]

Install new o-rings onto the male unions and install the unions into the rear turbocharger drain ports.

Torque Value:

Turbocharger Drain Port Unions 72 n•m [53 ft-lb]

Install new face seal o-rings and install the coolant return lines between the rear turbochargers and the right and left bank manifold reducers.

Torque Value:

Coolant Return Line 45 n•m [33 ft-lb]

Install the four p-clips onto the two hoses installed. Be sure that the hoses are **not** touching other adjacent hoses or engine components.

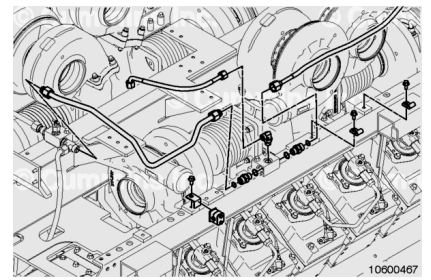
Torque Value:

P-clip Capscrew 45 n•m [33 ft-lb]

Install new o-rings onto the turbocharger supply line unions. Install the unions onto the turbochargers.

Torque Value:

Supply Line Union 72 n•m [53 ft-lb]



Install new o-rings onto the turbocharger supply line elbows. Install the elbows into the water manifold.

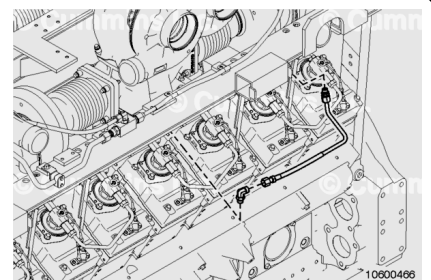
Torque Value:

Water Manifold Elbow 72 n•m [53 ft-lb]

Install new face seal o-rings and install the coolant supply lines on the four high pressure turbochargers.

Torque Value:

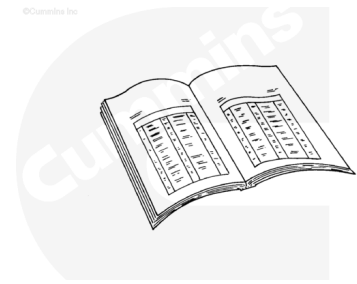
Coolant Supply Line 45 n•m [33 ft-lb]



Finishing Steps

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Operate the engine and check for leaks.

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